

Assignment 8: Time Series Analysis

Evan Lim

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
#Checking working directory  
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#Loading packages  
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    4.0.0      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
## as.Date, as.Date.numeric
```

```
library(trend)

#Setting theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "forestgreen"),
        panel.background = element_rect(fill="gray"),
        panel.grid.major = element_line(color="lightgray", linetype = "dotted"),
        legend.position = "top")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
#Importing datasets
ozone.2010 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"))
ozone.2011 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"))
ozone.2012 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"))
ozone.2013 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"))
ozone.2014 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"))
ozone.2015 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"))
ozone.2016 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"))
ozone.2017 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"))
ozone.2018 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"))
ozone.2019 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"))

#Combining into single dataframe
GaringerOzone <- rbind(ozone.2010,
                       ozone.2011,
                       ozone.2012,
                       ozone.2013,
                       ozone.2014,
                       ozone.2015,
                       ozone.2016,
                       ozone.2017,
```

```
ozone.2018,  
ozone.2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3 Setting date column as a date class  
GaringerOzone$Date <- mdy(GaringerOzone$Date)  
  
# 4 Wrangling to include specific columns  
GaringerOzone.wrangled <- GaringerOzone %>%  
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)  
  
# 5 Creating new data frame  
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),  
                          to = as.Date("2019-12-31"),  
                          by = "day"))  
  
colnames(Days) <- "Date"  
  
# 6 Combining the data frames  
GaringerOzone <- left_join(Days, GaringerOzone.wrangled)
```

```
## Joining with 'by = join_by(Date)'
```

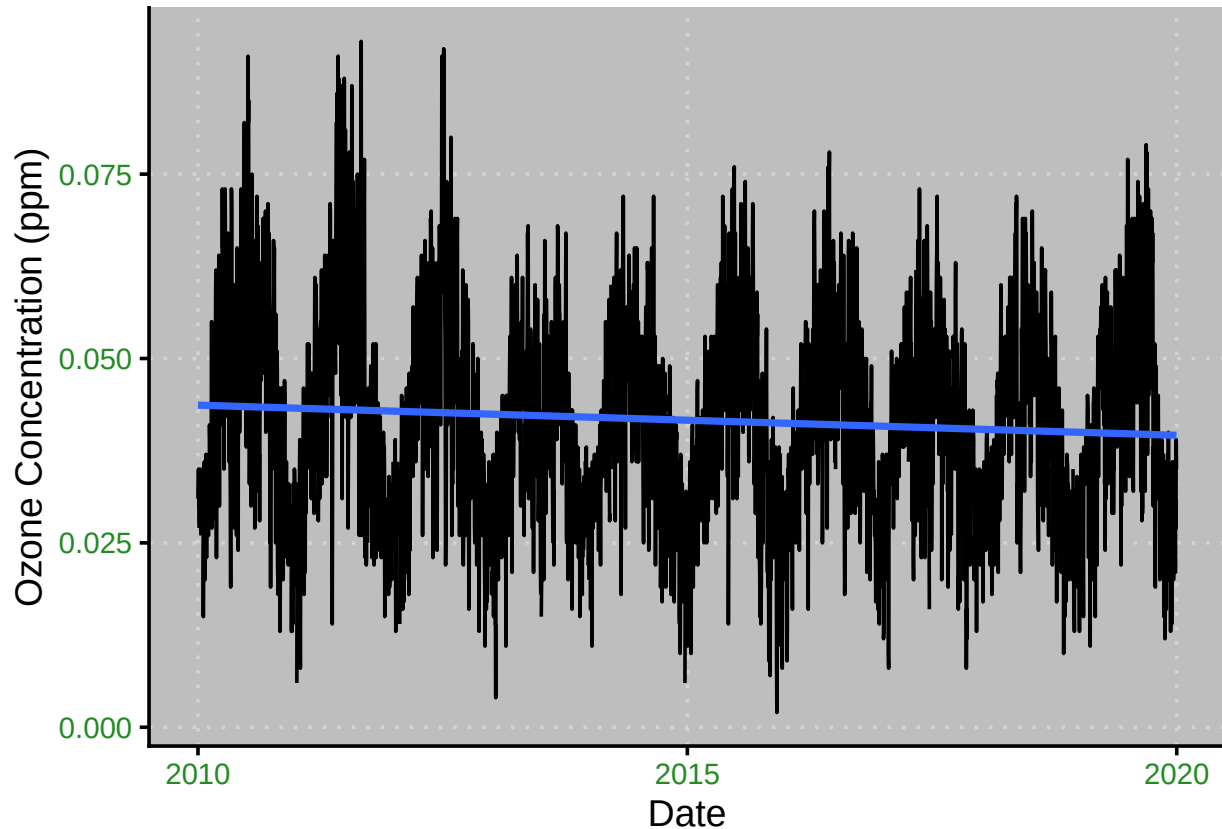
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7 Creating a line plot depicting ozone concentrations over time  
ggplot(GaringerOzone,  
  aes(x = Date,  
      y = Daily.Max.8.hour.Ozone.Concentration)) +  
  geom_line() +  
  geom_smooth(method = lm, se = FALSE) +  
  ylab("Ozone Concentration (ppm)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The plot does suggest a slight trend of ozone concentration decreasing over time from 2010 to 2020. Additionally, we can see seasonality here as well as there are up and down movements that are evenly distributed.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8 Using linear interpolation to replace NAs

```
GaringerOzone.clean <- GaringerOzone %>%
  mutate(Ozone.Clean = zoo::na.approx(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration))
```

Answer: Linear interpolation is the most simple of the methods and allows us to keep the seasonality of our data. Given that we have seasonal data and given that we don't want to use overly complex methods, linear interpolation is the method that makes the most sense here.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new `Date` column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9 Creating new data frame with aggregated data
```

```
GaringerOzone.monthly <- GaringerOzone.clean %>%  
  mutate(Month = month(Date),  
         Year = year(Date)) %>%  
  group_by(Year, Month) %>%  
  summarize(Mean_Ozone = mean(Ozone.Clean))
```

```
## 'summarise()' has grouped output by 'Year'. You can override using the  
## '.groups' argument.
```

```
GaringerOzone.monthly$Date <- my(paste0(GaringerOzone.monthly$Month,"-",GaringerOzone.monthly$Year))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

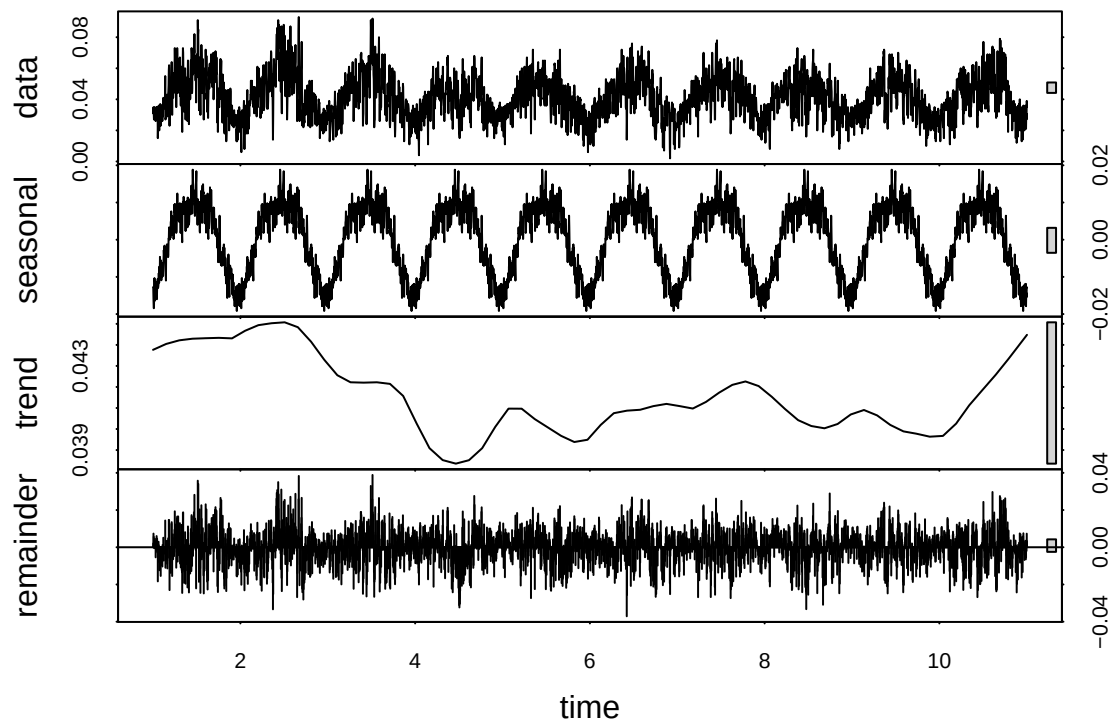
```
#10 Generating time series objects, one for daily and one for monthly
```

```
f_day <- day(first(GaringerOzone.clean$Date))  
f_month <- month(first(GaringerOzone.clean$Date))  
f_year <- year(first(GaringerOzone.clean$Date))  
  
GaringerOzone.daily.ts <- ts(GaringerOzone.clean$Ozone.Clean,  
                             start = c(f_day),  
                             frequency = 365)  
  
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Mean_Ozone,  
                               start = c(f_year,f_month),  
                               frequency = 12)
```

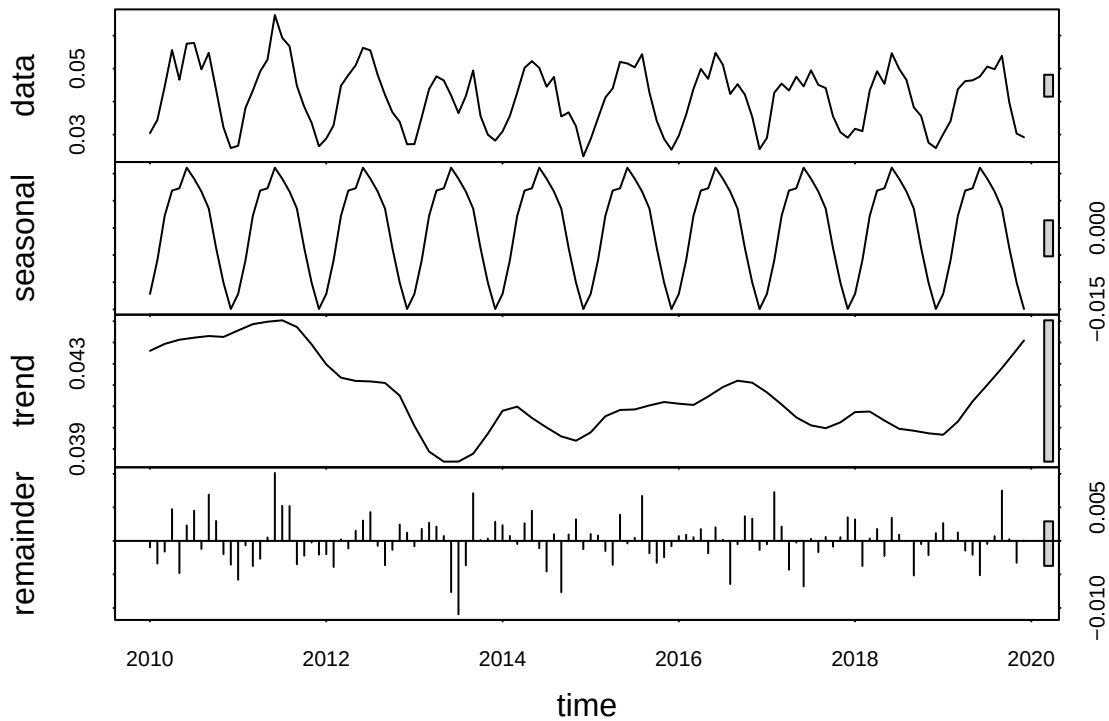
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11 Creating and plotting decompositions
```

```
GaringerOzone.daily.decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")  
GaringerOzone.monthly.decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")  
  
plot(GaringerOzone.daily.decomp)
```



```
plot(Garinger0zone.monthly.decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12 Running SMK test on monthly Ozone series
GaringerOzone.monthly.trend <- smk.test(GaringerOzone.monthly.ts)

GaringerOzone.monthly.trend
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
## S varS
## -77 1499
```

```
summary(GaringerOzone.monthly.trend)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
```

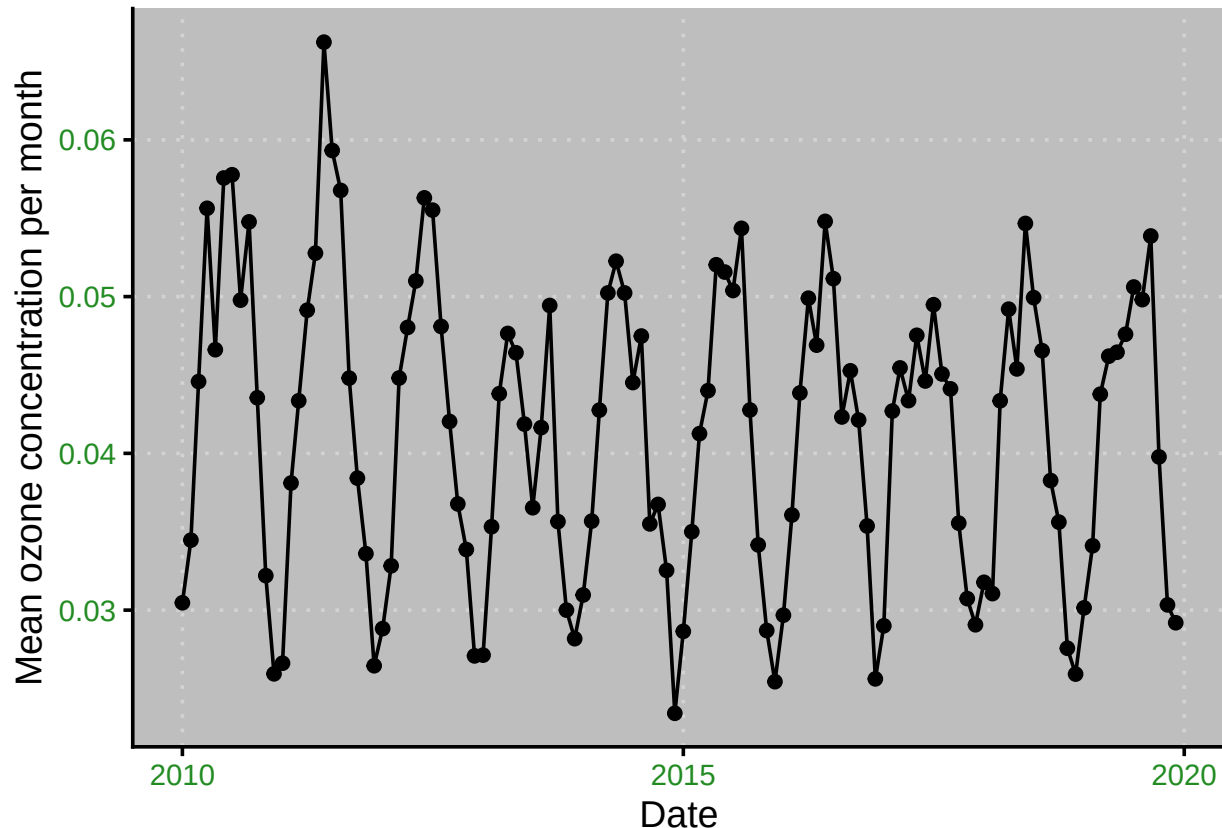
```
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S varS    tau      z Pr(>|z|)
## Season 1:  S = 0   15  125  0.333  1.252  0.21050
## Season 2:  S = 0   -1  125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4  124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 5:  S = 0  -15  125 -0.333 -1.252  0.21050
## Season 6:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 7:  S = 0  -11  125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7  125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10:  S = 0  -13  125 -0.289 -1.073  0.28313
## Season 11:  S = 0  -13  125 -0.289 -1.073  0.28313
## Season 12:  S = 0   11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: We use the seasonal Mann-Kendall test because our observational data is seasonal, and when trying to detect a trend, we need to account for seasonality. This is because seasonality can blur the detection of a trend due to the variations it causes in the data.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13 Creating a plot depicting mean monthly ozone concentrations over time

ggplot(GaringerOzone.monthly,
       aes(x = Date,
           y = Mean_Ozone)) +
  geom_point() +
  geom_line() +
  ylab("Mean ozone concentration per month")
```

14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: We conclude that ozone concentrations have changed over the 2010s at this station. The graph does indicate a slight decrease over the years as the peak values appear to drop slightly, and we can assume from the results of our Seasonal Mann-Kendall test that this trend is indeed statistically significant. Notably, our p-value is right at the border of statistical significance, so we might want to be mindful and perhaps conduct another analysis to confirm our results (p-value = 0.049).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15 Creating updated data frame without seasonal component
GaringerOzone.monthly_components <- as.data.frame(GaringerOzone.monthly.decomp$time.series)

GaringerOzone.monthly_components <- GaringerOzone.monthly_components %>%
  mutate(Observed = GaringerOzone.monthly$Mean_Ozone,
         Date = GaringerOzone.monthly$Date) %>%
  mutate(Minus_Seasonal = Observed - seasonal)
```

```

#Creating new time series object with new data frame
GaringerOzone.monthly.noseasonal.ts <- ts(GaringerOzone.monthly_components$Minus_Seasonal,
                                           start = c(f_year,f_month),
                                           frequency = 12)

#16 Running MK test on non-seasonal monthly Ozone series

GaringerOzone.monthly.noseasonal.trend <- mk.test(GaringerOzone.monthly.noseasonal.ts)

GaringerOzone.monthly.noseasonal.trend

##
## Mann-Kendall trend test
##
## data: GaringerOzone.monthly.noseasonal.ts
## z = -2.672, n = 120, p-value = 0.00754
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.179000e+03  1.943657e+05 -1.651376e-01

summary(GaringerOzone.monthly.noseasonal.trend)

##          Length Class  Mode
## data.name    1    -none- character
## p.value       1    -none- numeric
## statistic     1    -none- numeric
## null.value    1    -none- numeric
## parameter     1    -none- numeric
## estimates     3    -none- numeric
## alternative   1    -none- character
## method        1    -none- character
## pvalg         1    -none- numeric

```

Answer: After removing the seasonal component and running the Mann-Kendall test, we obtained a p-value of 0.007, in contrast to the seasonal Mann-Kendall test result of 0.049. This p-value is well below .05 and indicates greater statistical significance. This is likely due to the fact that seasonality in data can blur the trend itself, so in removing seasonality and examining just the trend with the Mann-Kendall test, we obtained a cleaner result. Thus, we can be more confident in stating there is a statistically significant downward trend in ozone concentrations over time at this site during the 2010s.